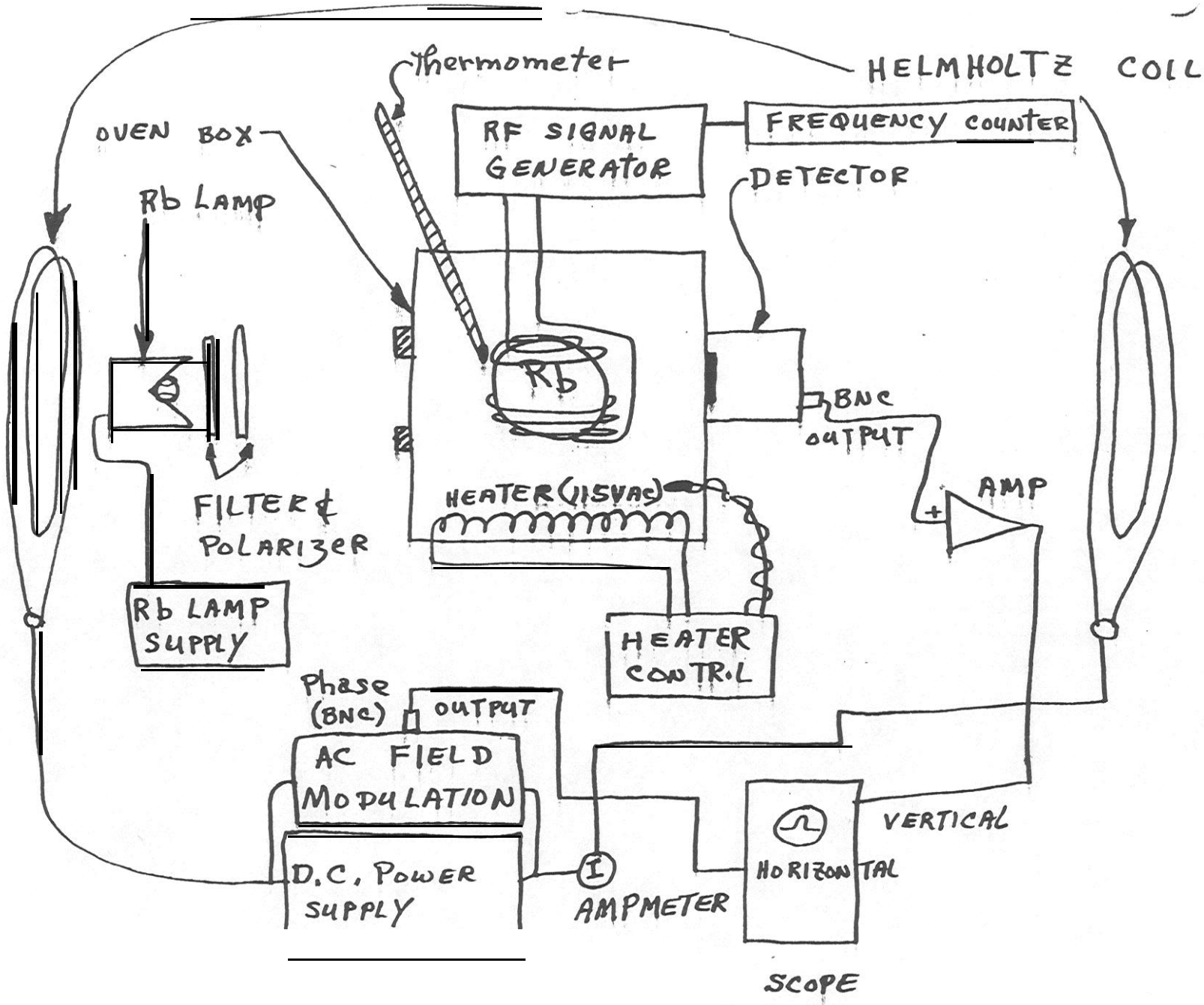
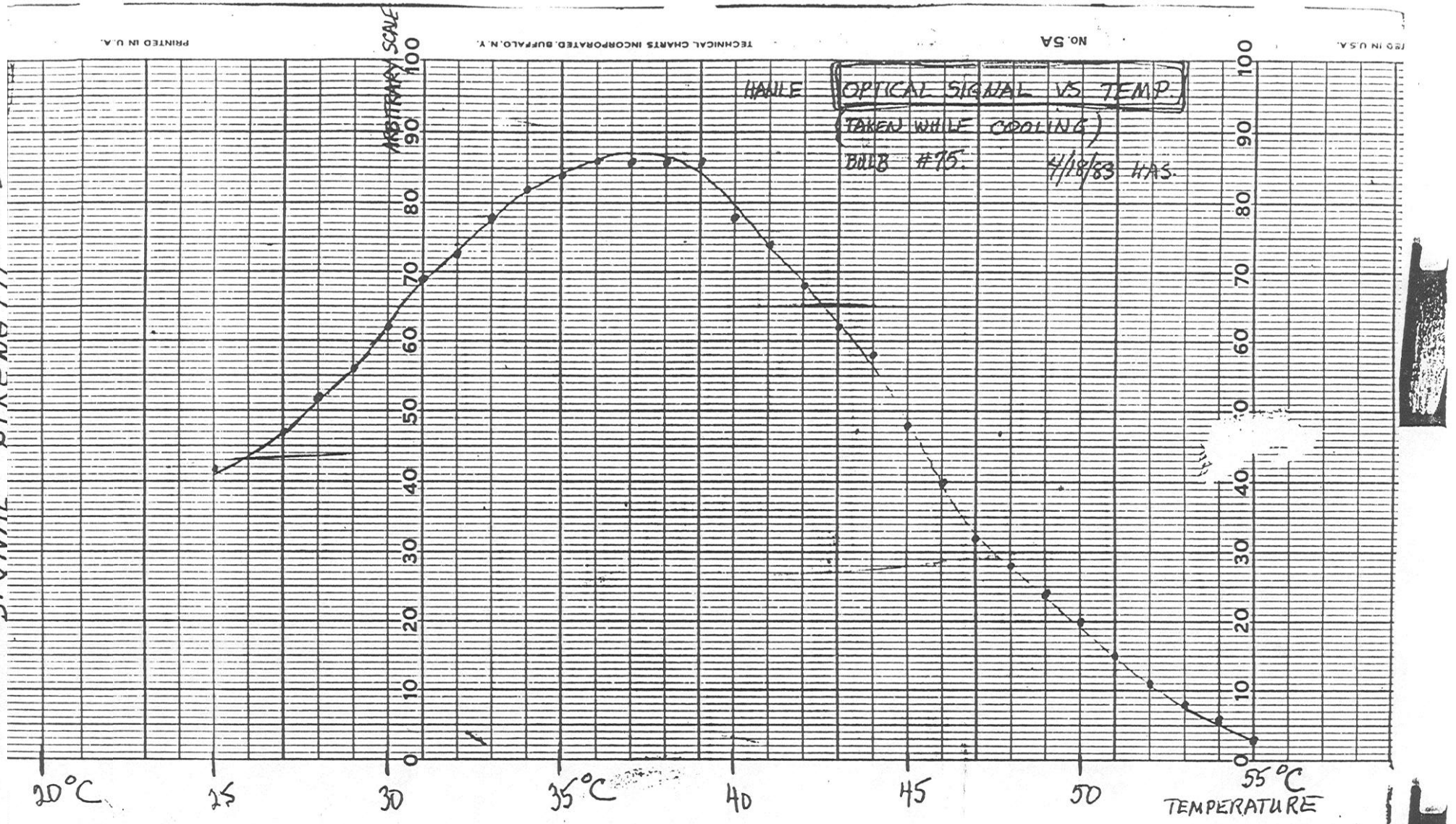


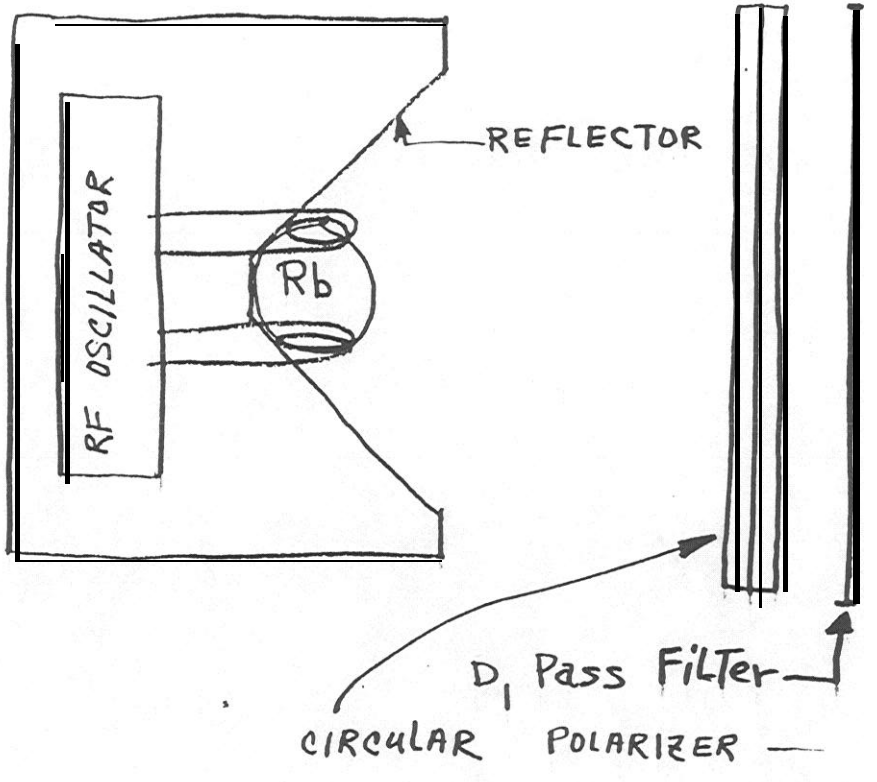
OPTICAL PUMPING BLOCK DIAGRAM:

-FIGURE 1-

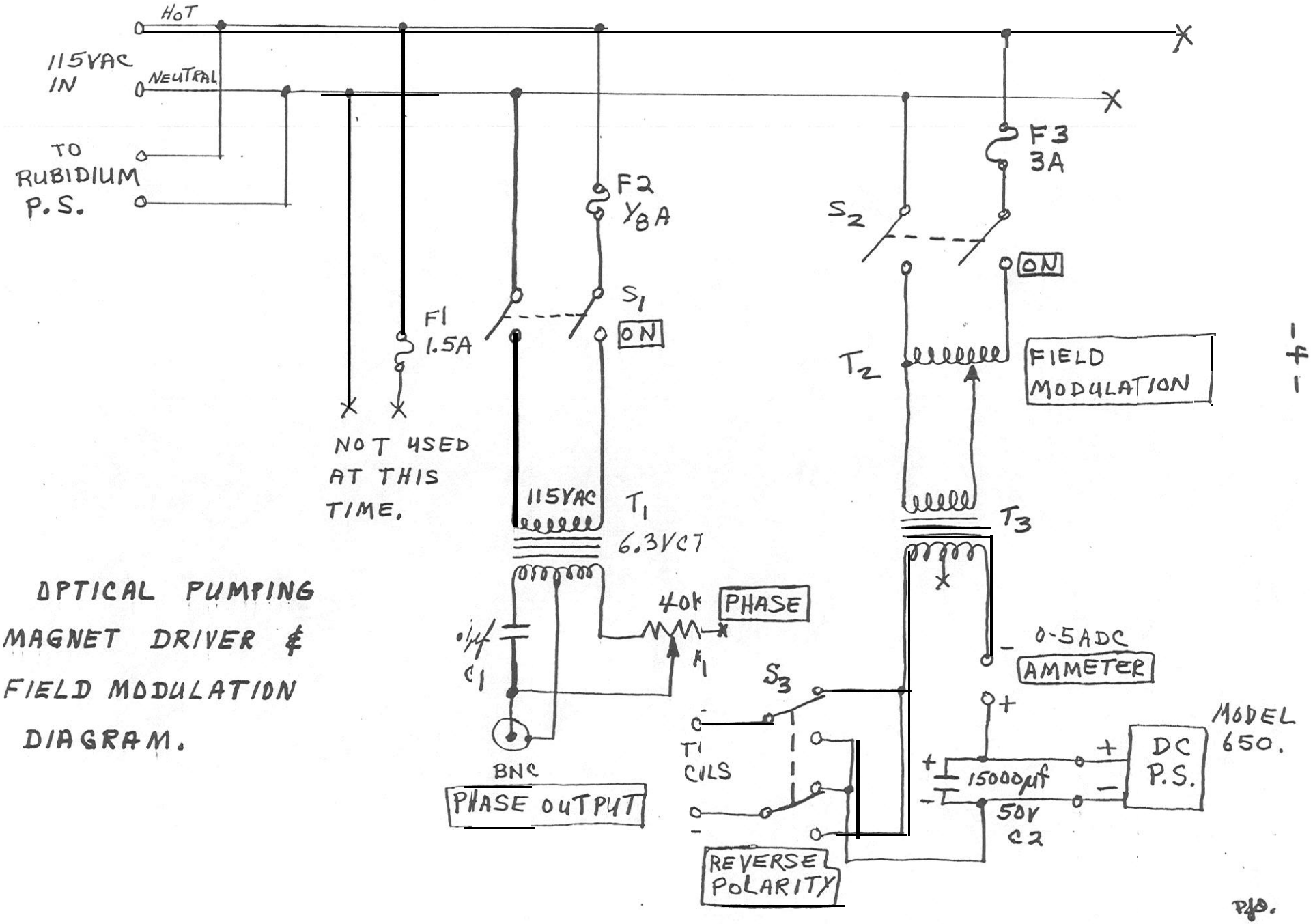


FOR Rb_{85}

Rb_{87} is HIGHER IN PEAK SIGNAL at A HIGHER TEMPERATURE.



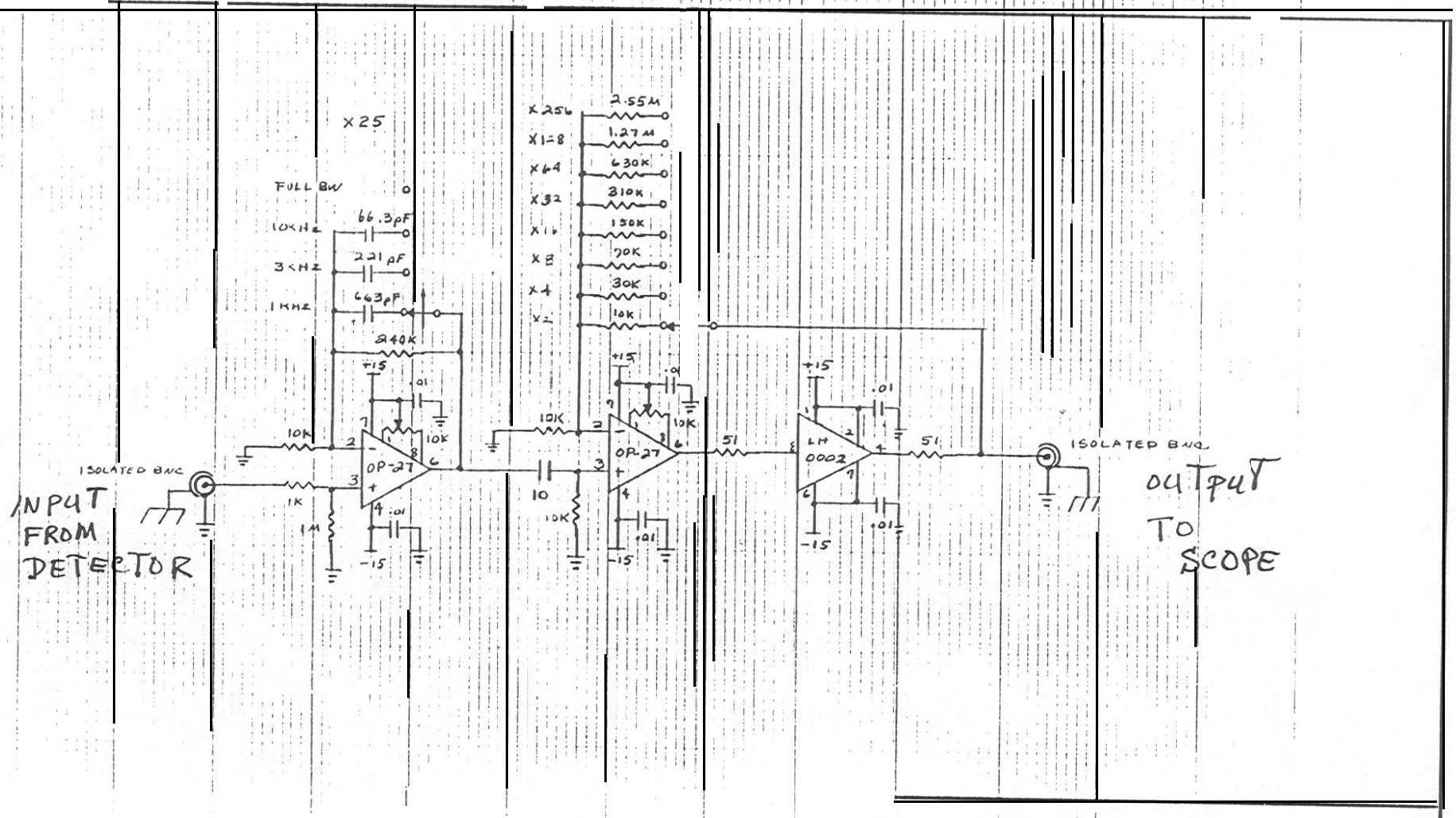
Rb LAMP ASSEMBLY



OPTICAL PUMPING MAGNET DRIVER & FIELD MODULATION DIAGRAM.

1 of 1

P.J.D.



PRE-AMP DIAGRAM

VOLTAGE AMPLIER

SCALE:	APPROVED BY:	DRAWN BY: Cummings
DATE: Oct 10, 84		REVISED

University of California Physics Dept

Electronics Shop

DRAWING NUMBER 82-56

11X17 PRINTED ON NO. 1000H-10 CLEARPRINT FADE-OUT

FALL 85-DJO

Photovoltaic Series

Planar Diffused Silicon Photodiodes



The Photovoltaic Detector series is utilized for applications requiring high sensitivity and moderate response speeds, with an additional sensitivity in the visible-blue region for the blue enhanced series. The spectral response ranges from 350 to 1100 nm, making the regular photovoltaic devices ideal for visible and near IR applications. For additional sensitivity in the 350 nm to 550 nm region, the blue enhanced devices are more suitable.

These detectors have high shunt resistance and low noise, and exhibit long term stability. Unbiased operation of these detectors offers stability under wide temperature variations in DC or low speed applications. For high light levels (greater than 10mW/cm2), the Photoconductive Series detectors should be considered for better linearity.

These detectors are not designed to be reverse biased! Very slight improvement in response time may be obtained with a slight bias. Applying a reverse bias of more than a few volts (>3V) will permanently damage the detectors. If faster response times are required, the Photoconductive Series should be considered.

Refer to the Photovoltaic Mode (PV) paragraph in the “Photodiode Characteristics” section of this catalog for detailed information on electronics set up.

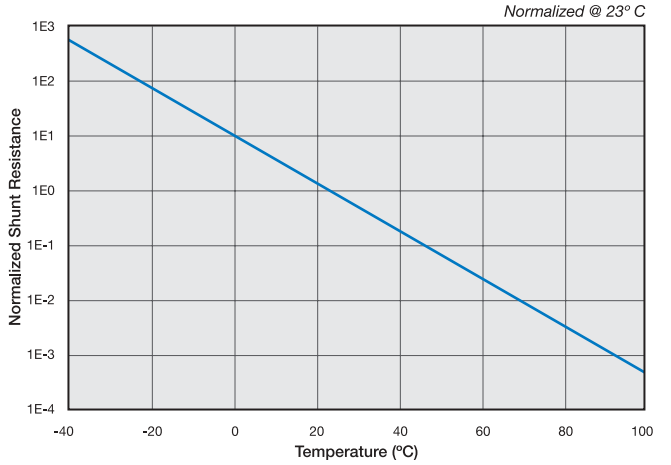
APPLICATIONS

- Colorimeters
- Photometers
- Spectroscopy Equipment
- Fluorescence

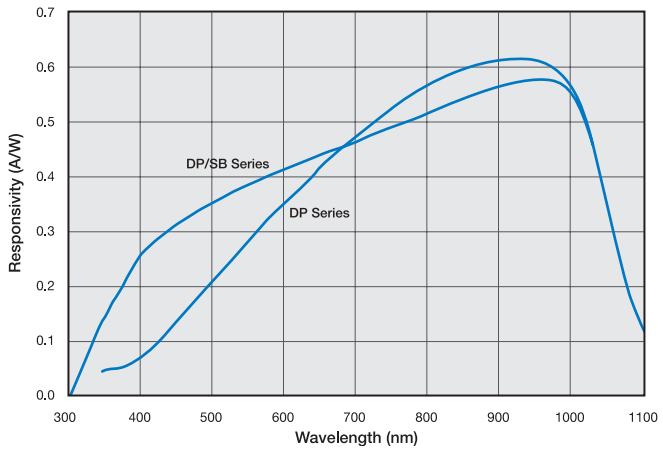
FEATURES

- Ultra Low Noise
- High Shunt Resistance
- Wide Dynamic Range
- Blue Enhanced

Typical Shunt Resistance vs. Temperature



Typical Spectral Response



Photovoltaic Series

Typical Electro-Optical Specifications at T_A=23ºC

Model Number	Active Area		Peak Responsivity Wavelength	Responsivity at λ_p		Capacitance (pF)	Shunt Resistance (GΩ)		NEP (W/√Hz)	Rise Time (ns)	Temp* Range (°C)		Package Style ¶
	Area (mm^2)	Dimension (mm)	λ_p nm	A/W		0 V	-10 mV		0V 970 nm	0 V 632 nm 50 Ω	Operating	Storage	
			typ.	min.	typ.	typ.	min.	typ.	typ.	typ.			
'DP' Series, Metal Package													
PIN-2DPI ‡	1.1	0.81 x 1.37	970	0.55	0.60	150	1.0	10	2.1 e -15	30	-40 ~ +100	-55 ~ +125	4 / TO-18
PIN-125DPL	1.6	1.27 sq.				160							8 / TO-18
PIN-3CDPI	3.2	1.27 x 2.54				320	0.5	5.0	3.0 e -15	50			4 / TO-18
PIN-3CDP													7 / TO-18
PIN-5DPI	5.1	2.54 ϕ				500	0.4	4.0	3.4 e -15	60			2 / TO-5
PIN-5DP													5 / TO-5
PIN-13DPI	13	3.6 sq				1200	0.35	3.5	3.6 e -15	150			2 / TO-5
PIN-13DP													5 / TO-5
PIN-6DPI	16.4	4.57 ϕ				2000	0.2	2.0	3.9 e -15	220			3 / TO-8
PIN-6DP													6 / TO-8
PIN-44DPI	44	6.6 sq				4300	0.1	1.0	4.8 e -15	475			3 / TO-8
PIN-44DP													6 / TO-8
PIN-10DPI	100	11.28 ϕ				9800	0.05	0.2	6.8 e -15	1000	-10 ~ +60	-20 ~ +70	10 / Lo-Prof
PIN-10DP													11 / BNC
PIN-25DP	613	27.9 ϕ				60000	0.002	0.1	3.0 e -14	6600			12 / BNC
'DP' Series, Plastic Package §													
FIL-3V	3.2	1.27 x 2.54	970	0.55	0.60	320	0.75	5.0	3.0 e -15	20	-10 ~ +60	-20 ~ +70	14 / Plastic
FIL-5V	5.1	2.54 ϕ				500	0.50	4.0	3.4 e -15	60			
FIL-20V	16.4	4.57 ϕ				2000	0.20	2.0	3.9 e -15	220			
FIL-44V	44	6.6 sq				4300	0.10	1.0	4.8 e -15	475			15 / Plastic
FIL-100V	100	11.28 ϕ				9800	0.05	0.2	6.8 e -15	1000			
PIN-220DP	200	10 x 20				20000	0.02	0.2	1.2 e -14	2200			27 / Plastic

Super Blue Enhanced 'DP/SB' Series (All Specifications @ λ= 410 nm, V_{BIAS}= 0V, R_L= 50Ω)

Model No.	Active Area/Dimension		Responsivity (A/W)		Capacitance (pF)	R_{sh} (MΩ)	NEP (W/√Hz)	Operating Current (mA)	Rise Time (µs)	Temp* Range (°C)		Package Style ¶
	mm^2	mm	min.	typ.	typ.	min.	typ.	max.	typ.	Operating	Storage	
PIN-040DP/SB	0.81	1.02 ϕ	0.15	0.20	60	1	2.0 e-14	0.5	0.02	-10 ~ +60	-20 ~ +70	1/ TO-18
PIN-5DP/SB †	5.1	2.54 ϕ			450	150	5.2 e -14	2.0	0.2			5 / TO-5
PIN-10DP/SB	100	11.28 ϕ			8800	10	2.0 e -13	10.0	2.0			11 / BNC
PIN-10DPI/SB												10 / Metal
PIN-220DP/SB	200	10 x 20			17000	5	2.9 e -13	10.0	4.0			27 / Plastic

‡ The 'I' suffix on the model number is indicative of the photodiode chip being isolated from the package by an additional pin connected to the case.
§ The photodiode chips in "FIL" series are isolated in a low profile plastic package. The have a large field of view as well as "in line" pins.
¶ For mechanical drawings please refer to pages 55 thru 66.
† Operating Temperature: -40 to +100 ºC, Storage Temperature: -55 to +125 ºC.
* Non-Condensing temperature and Storage Range, Non-Condensing Environment.

Photodiode Care and Handling Instructions

AVOID DIRECT LIGHT
Since the spectral response of silicon photodiode includes the visible light region, care must be taken to avoid photodiode exposure to high ambient light levels, particularly from tungsten sources or sunlight. During shipment from UDT Sensors, your photodiodes are packaged in opaque, padded containers to avoid ambient light exposure and damage due to shock from dropping or jarring.

AVOID SHARP PHYSICAL SHOCK
Photodiodes can be rendered inoperable if dropped or sharply jarred. The wire bonds are delicate and can become separated from the photodiode's bonding pads when the detector is dropped or otherwise receives a sharp physical blow.

CLEAN WINDOWS WITH OPTICAL GRADE CLOTH / TISSUE
Most windows on UDT Sensors photodiodes are either silicon or quartz. They should be cleaned with isopropyl alcohol and a soft (optical grade) pad.

OBSERVE STORAGE TEMPERATURES AND HUMIDITY LEVELS
Photodiode exposure to extreme high or low storage temperatures can affect the subsequent performance of a silicon photodiode. Storage temperature guidelines are presented in the photodiode performance specifications of this catalog. Please maintain a non-condensing environment for optimum performance and lifetime.

OBSERVE ELECTROSTATIC DISCHARGE (ESD) PRECAUTIONS
UDT Sensors photodiodes, especially with IC devices (e.g. Photops) are considered ESD sensitive. The photodiodes are shipped in ESD protective packaging. When unpacking and using these products, anti-ESD precautions should be observed.

DO NOT EXPOSE PHOTODIODES TO HARSH CHEMICALS
Photodiode packages and/or operation may be impaired if exposed to CHLOROTHENE, THINNER, ACETONE, or TRICHLOROETHYLENE.

INSTALL WITH CARE
Most photodiodes in this catalog are provided with wire or pin leads for installation in circuit boards or sockets. Observe the soldering temperatures and conditions specified below:

Soldering Iron:	Soldering 30 W or less Temperature at tip of iron 300°C or lower.	
Dip Soldering:	Bath Temperature: Immersion Time: Soldering Time:	260±5°C. within 5 Sec. within 3 Sec.
Vapor Phase Soldering:	DO NOT USE	
Reflow Soldering:	DO NOT USE	

Photodiodes in plastic packages should be given special care. Clear plastic packages are more sensitive to environmental stress than those of black plastic. Storing devices in high humidity can present problems when soldering. Since the rapid heating during soldering stresses the wire bonds and can cause wire to bonding pad separation, it is recommended that devices in plastic packages to be baked for 24 hours at 85°C.

The leads on the photodiode **SHOULD NOT BE FORMED**. If your application requires lead spacing modification, please contact UDT Sensors Applications group at (310)978-0516 before forming a product's leads. Product warranties could be voided.

Mechanical Drawings

Mechanical Specifications and Die Topography

1. Parameter Definitions:

A = Distance from top of chip to top of glass.
a = Photodiode Anode.
B = Distance from top of glass to bottom of case.
c = Photodiode Cathode
(Note: cathode is common to case in metal package products unless otherwise noted).
W = Window Diameter.
F.O.V. = Filed of View (see definition below).

2. Dimensions are in inches (1 inch = 25.4 mm).

3. Pin diameters are 0.018 ± 0.002" unless otherwise specified.

4. Tolerances (unless otherwise noted)

General: 0.XX ±0.01"
0.XXX ±0.005"
Chip Centering: ±0.010"
Dimension 'A': ±0.015"

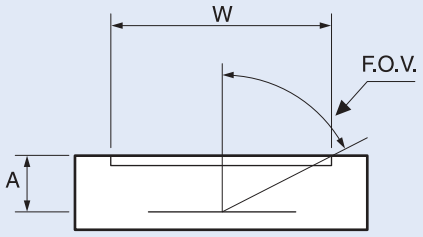
5. Windows

All **'UV'** Enhanced products are provided with QUARTZ glass windows, 0.027 ± 0.002" thick.

All **'XUV'** products are provided with removable windows.

All **'DLS'** PSD products are provided with A/R coated glass windows.

All **'FIL'** photoconductive and photovoltaic products are epoxy filled instead of glass windows.



$$F.O.V. = \tan^{-1}\left(\frac{W}{2A}\right)$$



For Further Assistance
Please Call One of Our Experienced
Sales and Applications Engineers

310-978-0516



- Or -
On the Internet at

www.udt.com

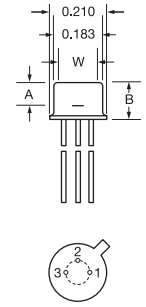
Mechanical Specifications

All units in inches. Pinouts are bottom view.

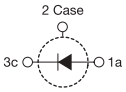
1 TO-18

Products:

PIN-020A
PIN-040A
InGaAs-1000
InGaAs-1500
PIN-040-DP/SB



Pin Circle Dia.=0.100

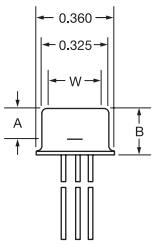


P/N	A	B	W
PIN-020A	0.075	0.200	0.155
PIN-040A	0.075	0.200	0.155
InGaAs -1000	0.055	0.150	0.080
InGaAs -1500	0.055	0.150	0.080

2 TO-5

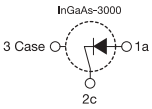
Products:

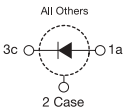
PIN-5DI
PIN-5DPI
PIN-13DI
PIN-13DPI
InGaAS-3000
PIN-5-YAG





Pin Circle Dia.=0.200



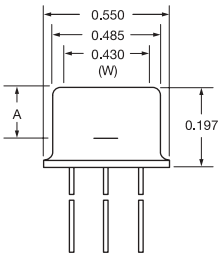


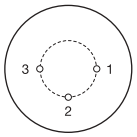
P/N	A	B	W
All Others	0.094	0.180	0.240
InGaAs -3000	0.077	0.185	0.177

3 TO-8

Products:

PIN-6DI
PIN-6DPI
PIN-44DI
PIN-44DPI





Pin Circle Dia.=0.295



PIN-6DI, PIN-6DPI



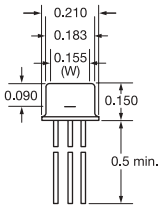
PIN-4DI, PIN-4DPI

P/N	A
PIN-6DI /6DPI	0.115
PIN-44DI /44DPI	0.125

4 TO-18

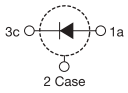
Products:

PIN-2DI
PIN-2DPI
PIN-3CDP
PIN-3CDPI





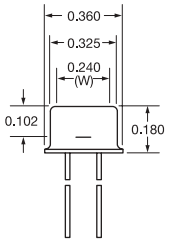
Pin Circle Dia.=0.100



5 TO-5

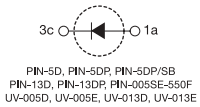
Products:

PIN-5D
PIN-5DP
PIN-5DP/SB
PIN-13D
PIN-13DP
PIN-005E-550F
UV-001
UV-005
UV-005D
UV-005E
UV-013D
UV-013E
UV-015





Pin Circle Dia.=0.200



PIN-5D, PIN-5DP, PIN-5DP/SB
PIN-13D, PIN-13DP, PIN-005SE-550F
UV-005D, UV-005E, UV-013D, UV-013E

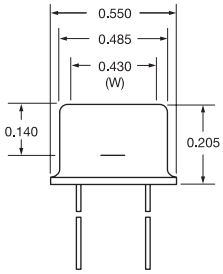


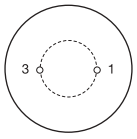
UV-001, UV-005, UV-015

6 TO-8

Products:

PIN-6D
PIN-6DP
PIN-44D
PIN-44DP
UV-020
UV-035D
UV-035E
UV-035





Pin Circle Dia.=0.295



PIN-6D, PIN-6DP, PIN-44D, PIN-44DP
UV-020, UV-035D, UV-035E

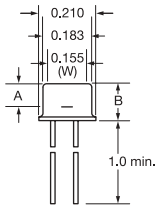


UV-035

7 TO-18

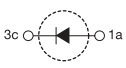
Products:

PIN-3CD
PIN-3CDP
BPX-65





Pin Circle Dia.=0.100

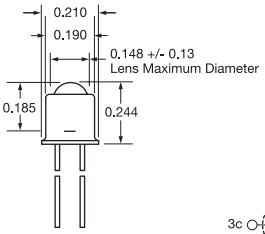


P/N	A	B
PIN-3CD / 3CDP	0.087	0.146
BPX-65	0.075	0.200

8 TO-18

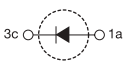
Products:

PIN-125DPL





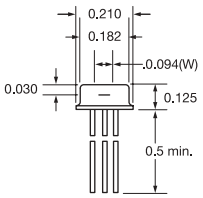
Pin Circle Dia.=0.100



9 TO-18

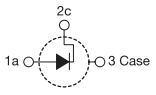
Products:

PIN-HR005
PIN-HR008
PIN-HR020
PIN-HR026
PIN-HR040





Pin Circle Dia.=0.100

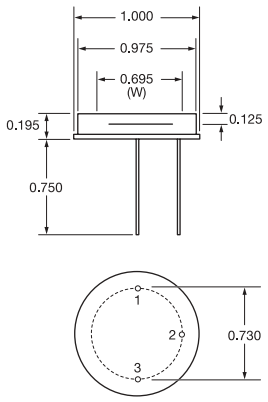


All units in inches. Pinouts are bottom view.

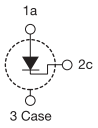
10 Low Profile

Products:

PIN-10DI
PIN-10DPI
PIN-10DPI/SB
UV-50L
UV-100L



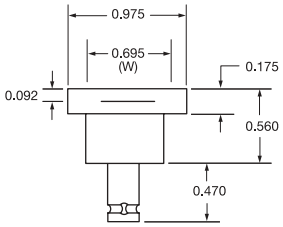
Pin Circle Dia.=0.100



11 BNC

Products:

PIN-10D
PIN-10DP
PIN-10DP/SB
UV-50
UV-100

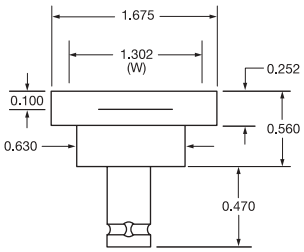


Outer Contact — Anode	PIN-10D, PIN-10DP, PIN-10DP/SB
Outer Contact — Cathode	UV-50, UV-100

12 BNC

Products:

PIN-25D
PIN-25DP

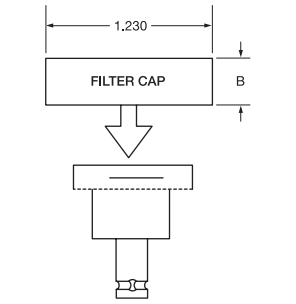


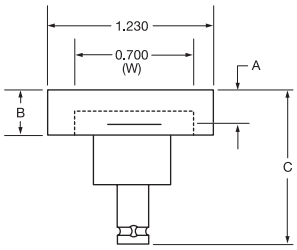
Outer Contact — Anode

13 Special BNC

Products:

PIN-10AP
PIN-10DF





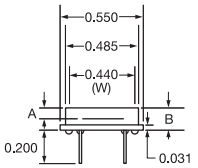
Dimensions

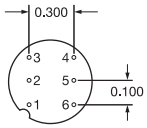
P/N	A	B	C
PIN-10DF	0.217	0.330	1.020
PIN-10AP	0.386	0.550	1.415

14 Special Plastic

Products:

FIL-3C
FIL-5C
FIL-20C
FIL-3V
FIL-5V
FIL-20V
FIL-UV20
FIL-UV005
FIL-S2DG
FIL-S4DG
FIL-C4DG





Dimensions

P/N	A	B
FIL-3V(C) FIL-5V(C) FIL-20V(C)	0.060	0.130
FIL-UV005 FIL-UV20 FIL-S4DG FIL-C4DG FIL-S2DG	0.087	0.152

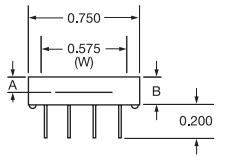
Pinouts

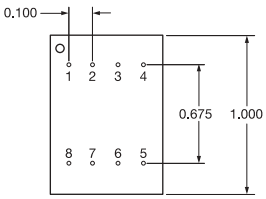
P/N	1	2	3	4	5	6
FIL-3V(C) FIL-5V(C) FIL-20V(C) FIL-UV005 FIL-S2DG	a	-	c	a	-	c
FIL-UV20	c	-	a	c	-	a
FIL-S4DG	a	a	c	a	a	c
FIL-C4DG	c	a	c	c	a	c

15 Special Plastic

Products:

FIL-44C
FIL-100C
FIL-44V
FIL-100V
FIL-UV50
FIL-UV100
FIL-S9DG
FIL-C10DG





Dimensions

P/N	A	B
FIL-44V(C) FIL-100V(C)	0.052	0.130
FIL-UV50 FIL-UV100	0.090	0.155
FIL-S9DG	0.072	0.155
FIL-C10DG	0.082	0.155

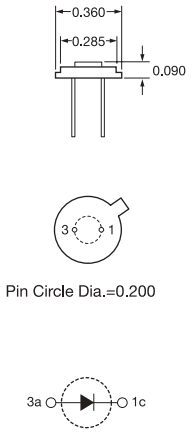
Pinouts

P/N	1	2	3	4	5	6	7	8
FIL-44V(C) FIL-100V(C)	a	-	-	c	a	-	-	c
FIL-UV50 FIL-UV100	c	-	-	a	c	-	-	a
FIL-S9DG	a	c	c	a	a	c	c	a
FIL-C10DG	c	a	a	c	c	a	a	c

All units in inches. Pinouts are bottom view.

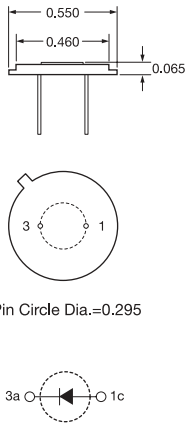
22 TO-5

Products:
XUV-005



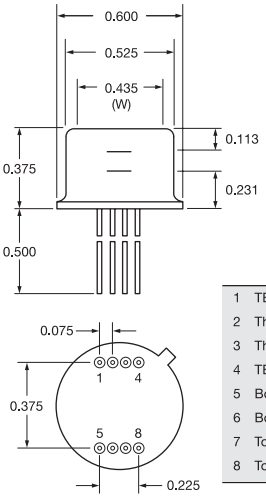
23 TO-8

Products:
XUV-020
XUV-035



24 TO-8

Products:
PIN-DSIn-TEC

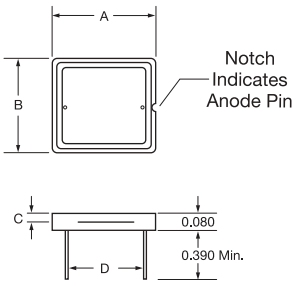


Pinout

1	TEC (-)
2	Thermistor
3	Thermistor
4	TEC (+)
5	Bottom InGaAs, Cathode
6	Bottom InGaAs, Anode
7	Top Silicon, Anode
8	Top Silicon, Cathode

25 Special Ceramic / Plastic

Products:
RD-100
RD-100A
UV-35P
UV-005EC
UV-035EC
UV-100EC
UV-005DC
UV-035DC
UV-100DC
XUV-50C
XUV-100C

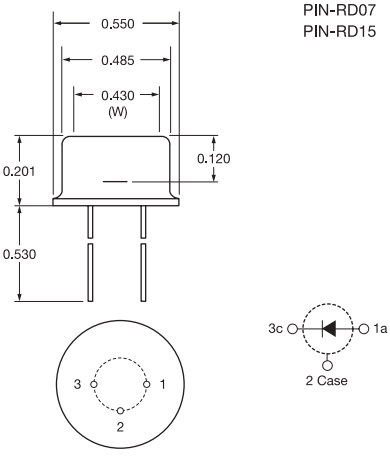


Dimensions

P/N	A	B	C	D
UV-005EC	0.400	0.350	0.030	0.180
UV-035EC	0.400	0.350	0.030	0.290
UV-100EC	0.650	0.590	0.048	0.500
UV-005DC	0.400	0.350	0.030	0.180
UV-035DC	0.400	0.350	0.030	0.290
UV-100DC	0.650	0.590	0.053	0.500
XUV-50C	0.490	0.590	0.027	0.490
XUV-100C	0.490	0.590	0.027	0.490
RD-100	0.650	0.590	0.027	0.490
RD-100A	0.650	0.590	0.027	0.490
UV-35P	0.390	0.345	0.050	0.275

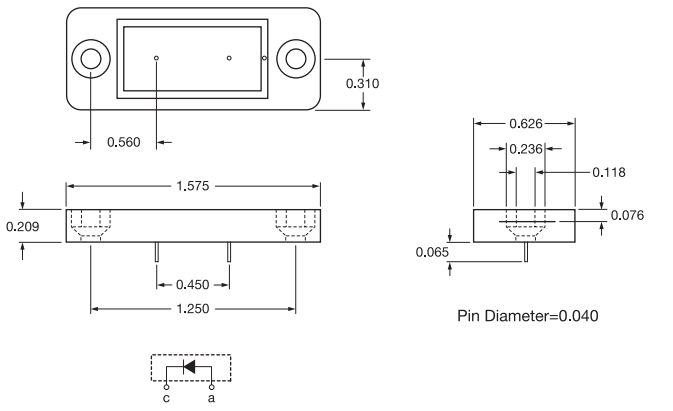
26 TO-8

Products:
PIN-RD07
PIN-RD15



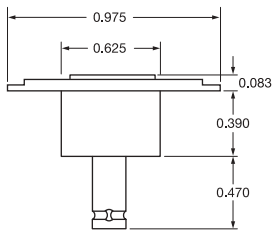
27 Special Plastic

Products:
PIN-220D
PIN-220DP
PIN-220DP/SB



28 BNC

Products:
XUV-100



BNC Connector
Outer Contact = Cathode